Codio Project File Edit Find View Tools Education Help Configure... Project Index (static) Configure... SANDIPAN-DEY

Terminal

```
codio@kiwivatican-brownsunday:~/workspace$
```

Guide

Collapse

Examples Module

The default size of a pixel art image is seven pixels by seven pixels. The `house` function provides an example of pixel art with the default dimensions. Instantiate an `Image` object and add each row of pixels. Use the `display()` method to draw the image to the terminal

```
def house():
  img = pixels.Image()
  img.add_row(0, 'cyan')
  img.add_row(1, 'cyan', 'yellow', 'cyan', 'red', 'cyan', 'cyan', 'cyan')
  img.add_row(2, 'cyan', 'cyan', 'red', 'red', 'red', 'cyan', 'cyan')
  img.add_row(3, 'cyan', 'red', 'red', 'red', 'red', 'red', 'cyan')
  img.add_row(4, 'cyan', 'white', 'white', 'white', 'white', 'white', 'cyan')
  img.add_row(5, 'cyan', 'white', 'white', 'black', 'white', 'white', 'cyan')
  img.add_row(6, 'green')
  img.display()
```

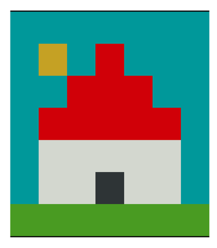
Add the code below to the testing section.

```
# test your code here
if __name__ == '__main__':
  house()
```

Run the `examples` module to test the `house` function in the **bottom** panel.

```
python3 examples.py
```

You should see the following output:



Example Notes

The final example is musical notes. This image is done with custom dimensions so the image is more intricate. Create the `notes` function, which has no parameters. Instantiate an `Image` object with a width of 16 pixels and a height of 14 pixels. Add each row of pixels, and use `display()` to draw it to the terminal.

```
def notes():
  img = pixels.Image(16, 14)
  img.add_row(0, 'magenta')
  img.add_row(1, 'magenta', 'magenta', 'magenta', 'magenta',
                 'magenta', 'magenta', 'black', 'black',
                 'black', 'black', 'black', 'black',
                 'black', 'black', 'black', 'magenta')
  img.add_row(2, 'magenta', 'magenta', 'magenta', 'magenta',
                 'magenta', 'black', 'white', 'white',
                 'white', 'white', 'white', 'white',
                 'white', 'white', 'black', 'magenta')
  img.add_row(3, 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'white', 'white',
                 'white', 'white', 'white', 'white',
                 'white', 'white', 'black', 'magenta')
  img.add_row(4, 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'black',
                 'black', 'black', 'black', 'black',
                 'black', 'white', 'black', 'magenta')
  img.add_row(5, 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'magenta',
                 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'magenta')
  img.add_row(6, 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'magenta',
                 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'magenta')
  img.add_row(7, 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'magenta',
                 'magenta', 'magenta', 'magenta', 'magenta',
                 'black', 'white', 'black', 'magenta')
  img.add_row(8, 'magenta', 'magenta', 'black', 'black',
                 'black', 'white', 'black', 'magenta',
                 'magenta', 'magenta', 'black', 'black',
                 'black', 'white', 'black', 'magenta')
  img.add_row(9, 'magenta', 'black', 'white', 'white',
                 'white', 'white', 'black', 'magenta',
                 'magenta', 'black', 'white', 'white',
                 'white', 'white', 'black', 'magenta')
  img.add_row(10, 'magenta', 'black', 'white', 'white',
                  'white', 'white', 'black', 'magenta',
                  'magenta', 'black', 'white', 'white',
                  'white', 'white', 'black', 'magenta')
  img.add_row(11, 'magenta', 'black', 'white', 'white',
                  'white', 'white', 'black', 'magenta',
                  'magenta', 'black', 'white', 'white',
                  'white', 'white', 'black', 'magenta')
  img.add_row(12, 'magenta', 'magenta', 'black', 'black',
                  'black', 'black', 'magenta', 'magenta',
                  'magenta', 'magenta', 'black', 'black',
                  'black', 'black', 'magenta', 'magenta')
  img.add_row(13, 'magenta')
  img.display()
```

Add the code below to the testing section.

```
# test your code here
if __name__ == '__main__':
  notes()
```

Run the `examples` module to test the `notes` function in the **bottom** panel.

```
python3 examples.py
```

You should see the following output:

Once we have tested all of the example images, we need to change the import statement so everything works when we convert it to a wheel. Your import statement should now look like this:

```
import pixel_art.pixels as pixels
```

▶ **Code**

Lab Question

How do you install both the `requests` and `numpy` packages with `pip`? **Hint**, there is more than one correct answer.

- [ ]
```
python3 -m pip requests numpy
```

- [ ]
```
python3 -m pip install requests
python3 -m pip install numpy
```

- [ ]
```
python3 -m pip install requests numpy
```

- [ ]
```
python3 -m pip install requests, numpy
```

The correct answers are:

```
python3 -m pip install requests
python3 -m pip install numpy
```

You can use the same `pip` command twice to install each package individually.

```
python3 -m pip install requests numpy
```

Or, you can install both packages with a single command. Notice that only a space (not a comma) separates the two packages. You must also use the `install` command with `pip`.

Check It!

Next